EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 08

Statistical Inference

1 Concepts | 50 questions

B5 Landscape | one question per teaching page

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Chapter 08 | Statistical Inference

CLASSROOM ROUTE

Concept 1 | Fundamentals of Random Variables

50 questions

Official lessons: 8-1, 8-2, 8-3, 8-4, 8-5, 8-6, 8-7

The Formula Bank changes at each Concept boundary.

Concept 1 | Fundamentals of Random Variables OFFICIAL LESSONS 8-1, 8-2, 8-3, 8-4, 8-5, 8-6, 8-7

Fundamentals of Random Variables

Equation-first recall | use the same rail beside every question in this Concept

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Start with the family cue, write the first move, then use the working grid.

Concept 1 | Fundamentals of Random Variables

F01 | V01

Q001 | WRITTEN

C08-K01-8-1-WU

For the sum of two dice, construct the probability distribution and find P5 <= X <= 8.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum x P(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2 V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | V02

Q002 | WRITTEN

C08-K01-8-1-TRY

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | V03

Q003 | WRITTEN

C08-K01-8-1-EX

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | V01

Q004 | WRITTEN C08-K01-8-2-WU

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | V02

Q005 | WRITTEN C08-K01-8-2-TRY

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | V03

Q006 | WRITTEN

C08-K01-8-2-EX

Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | V01

Q007 | WRITTEN

C08-K01-8-3-WU

For tails in four fair coins, find E(3X+2), E(-4X+1), and EX^2.

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | V02

Q008 | WRITTEN

C08-K01-8-3-TRY

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | V03

Q009 | WRITTEN

C08-K01-8-3-EX

Solve both source Exercise groups of transformed expectations for coins and for balls.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | V01

Q010 | WRITTEN

C08-K01-8-4-WU

For black balls drawn from the stated box, find EX and VX.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | V02

Q011 | WRITTEN

C08-K01-8-4-TRY

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | V03

Q012 | WRITTEN

C08-K01-8-4-EX

Find EX and VX for all five source Exercise random variables.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F05 | V01

Q013 | WRITTEN

C08-K01-8-5-WU

Find the standard deviation of tails when two biased coins have P(head)=4/7 and P(tail)=3/7.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum x P(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2 V(X)$$

Concept 1 | Fundamentals of Random Variables

F05 | V02

Q014 | WRITTEN C08-K01-8-5-TRY

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F05 | V03

Q015 | WRITTEN

C08-K01-8-5-EX

Find the standard deviation for both source probability-table exercises.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | V01

Q016 | WRITTEN

C08-K01-8-6-WU

For the number of winning tickets in the source draw, find VY and sigma$Y$ for Y=2X+1 and Y=-3X-2.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X=x)=1$$

Mean

$$E(X)=\sum xP(X=x)$$

Variance

$$V(X)=E(X^2)-[E(X)]^2$$

Standard deviation

$$\sigma(X)=\sqrt{V(X)}$$

Transformation

$$E(aX+b)=aE(X)+b \quad V(aX+b)=a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | V02

Q017 | WRITTEN

C08-K01-8-6-TRY

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | V03

Q018 | WRITTEN

C08-K01-8-6-EX

For a single die, find VY and sigma$Y$ for Y=X+2, Y=3X+1 and Y=-2X+4.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum x P(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2 V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | V01

Q019 | WRITTEN

C08-K01-8-7-WU

Solve the two source transformation problems: a net prize after an entry fee, and finding a, b from E(aX+b)=7/3 and sigma(aX+b)=3.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | V02

Q020 | WRITTEN

C08-K01-8-7-TRY

Solve the two source Try problems involving net winnings and determining a, b for a die transformation.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X=x)=1$$

Mean

$$E(X)=\sum xP(X=x)$$

Variance

$$V(X)=E(X^2)-[E(X)]^2$$

Standard deviation

$$\sigma(X)=\sqrt{V(X)}$$

Transformation

$$E(aX+b)=aE(X)+b \quad V(aX+b)=a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | V03

Q021 | WRITTEN

C08-K01-8-7-EX

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | C08-K01-F01-VMCQ

Q022 | MCQ

C08-K01-8-1-GM01

A probability distribution must have probabilities that sum to Which of the following is correct?:

A 0

B 1

C the number of outcomes

D the mean

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X=x)=1$$

Mean

$$E(X)=\sum xP(X=x)$$

Variance

$$V(X)=E(X^2)-[E(X)]^2$$

Standard deviation

$$\sigma(X)=\sqrt{V(X)}$$

Transformation

$$E(aX+b)=aE(X)+b \quad V(aX+b)=a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | C08-K01-F01-VMCQ

Q023 | MCQ · C08-K01-8-1-GM02

For two fair dice, $PX = 2$ is:

- **A** 1/6
- **B** 1/36
- **C** 2/36
- **D** 0

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | C08-K01-F01-VMCQ

Q024 | MCQ

C08-K01-8-1-GM03

The number of heads in three coins can be Which of the following is correct?:

A 1,2,3

B 0,1,2,3

C 0,3 only

D any integer

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | C08-K01-F02-VMCQ

Q025 | MCQ

C08-K01-8-2-GM04

The expected value is calculated as:

A sum P$X = x$

B sum xP$X = x$

C largest x

D variance squared

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | C08-K01-F02-VMCQ

Q026 | MCQ

C08-K01-8-2-GM05

The mean number of tails in four fair coins is:

A 1	**B** 2
C 3	**D** 4

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | C08-K01-F02-VMCQ

Q027 | MCQ

C08-K01-8-2-GM06

For a fair die, EX is:

A 3

B 3.5

C 4

D 6

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | C08-K01-F03-VMCQ

Q028 | MCQ

C08-K01-8-3-GM07

E(aX+b) equals Which of the following is correct?:

A aEX

B aEX+b

C EX+b only

D a+b

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | C08-K01-F03-VMCQ

Q029 | MCQ

C08-K01-8-3-GM08

In general, EX^2 is:

A $E[X]^2$

B equal to EX

C a weighted square sum

D always zero

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | C08-K01-F03-VMCQ

Q030 | MCQ C08-K01-8-3-GM09

If EX=2, then E(3X+2) is:

A 6

B 8

C 10

D 12

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | C08-K01-F04-VMCQ

Q031 | MCQ

C08-K01-8-4-GM10

The variance identity is:

A $E[X] - E[X^2]$

B $E[X^2] - E[X]^2$

C $E[X]^2$

D sqrt V

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | C08-K01-F04-VMCQ

Q032 | MCQ

C08-K01-8-4-GM11

Variance can never be Which of the following is correct?:

A positive

B zero

C negative

D fractional

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | C08-K01-F04-VMCQ

Q033 | MCQ

C08-K01-8-4-GM12

The first step in a variance problem is to Which of the following is correct?:

A draw the distribution

B take a square root

C ignore probabilities

D add the values

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F05 | C08-K01-F05-VMCQ

Q034 | MCQ

C08-K01-8-5-GM13

Standard deviation is:

A VX

B sqrt$V(X)$

C VX squared

D EX

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X=x)=1$$

Mean

$$E(X)=\sum xP(X=x)$$

Variance

$$V(X)=E(X^2)-[E(X)]^2$$

Standard deviation

$$\sigma(X)=\sqrt{V(X)}$$

Transformation

$E(aX+b)=aE(X)+b \quad V(aX+b)=a^2V(X)$

Concept 1 | Fundamentals of Random Variables

F05 | C08-K01-F05-VMCQ

Q035 | MCQ

C08-K01-8-5-GM14

Standard deviation has units Which of the following is correct?:

A squared units

B the same units as X

C no units

D units cubed

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum x P(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | C08-K01-F06-VMCQ

Q036 | MCQ

C08-K01-8-6-GM15

V(aX+b) equals Which of the following is correct?:

A aVX+b

B a squared times VX

C |a|VX

D VX+b

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | C08-K01-F06-VMCQ

Q037 | MCQ C08-K01-8-6-GM16

sigma(aX+b) equals Which of the following is correct?:

A a sigma$X$

B |a| sigma$X$

C a squared sigma$X$

D sigma$X$+b

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | C08-K01-F06-VMCQ

Q038 | MCQ

C08-K01-8-6-GM17

For Y=X+3, the variance is:

A VX+3

B VX

C 3VX

D 9VX

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | C08-K01-F07-VMCQ

Q039 | MCQ C08-K01-8-7-GM18

A net prize after a fixed fee is modeled as Which of the following is correct?:

A Y=X+b

B Y=aX+b

C $Y = X^2$

D Y=a+b only

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | C08-K01-F07-VMCQ

Q040 | MCQ

C08-K01-8-7-GM19

If a>0 and sigma(aX+b)=3, then a is:

A 3 sigma$X$

B 3/sigma$X$

C sigma$X$/3

D 3+b

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | C08-K01-F07-VMCQ

Q041 | MCQ C08-K01-8-7-GM20

Changing b in aX+b changes Which of the following is correct?:

A variance only

B standard deviation only

C the mean only

D nothing

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F04 | C08-K01-F04-VMCQ

Q042 | MCQ   C08-K01-8-4-GM21

If EX=2 and EX^2=7, then VX is:

A 3

B 5

C 7

D 9

WORKING AREA

FORMULA BANK

Distribution
$$\sum P(X = x) = 1$$

Mean
$$E(X) = \sum xP(X = x)$$

Variance
$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation
$$\sigma(X) = \sqrt{V(X)}$$

Transformation
$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | C08-K01-F02-VQUIZ

Quiz W1 | QUIZ WRITTEN

C08-K01-Q-W01

Find EX and VX for the number of heads in three fair coins.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | C08-K01-F06-VQUIZ

Quiz W2 | QUIZ WRITTEN C08-K01-Q-W02

For Y=4X-1 with VX=2, find VY and sigma$Y$.

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F03 | C08-K01-F03-VQUIZ

Quiz W3 | QUIZ WRITTEN C08-K01-Q-W03

For a fair die, find E(2X+5).

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | C08-K01-F07-VQUIZ

Quiz W4 | QUIZ WRITTEN C08-K01-Q-W04

If sigma$X$=2 and EX=3, find sigma(-5X+7).

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F06 | C08-K01-F06-VQUIZ

Quiz M1 | QUIZ MCQ

C08-K01-Q-M05

If VX=5, V(3X+1) is:

A 5

B 8

C 15

D 45

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F02 | C08-K01-F02-VQUIZ

Quiz M2 | QUIZ MCQ

C08-K01-Q-M06

For a fair die EX is:

A 2.5

B 3

C 3.5

D 4

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F01 | C08-K01-F01-VQUIZ

Quiz M3 | QUIZ MCQ

C08-K01-Q-M07

A probability table total must be Which of the following is correct?:

A 0

B 1

C 2

D the mean

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

Concept 1 | Fundamentals of Random Variables

F07 | C08-K01-F07-VQUIZ

Quiz M4 | QUIZ MCQ

C08-K01-Q-M08

If sigma$X$=4, sigma(X-9) is:

A 0

B 4

C 13

D 16

WORKING AREA

FORMULA BANK

Distribution

$$\sum P(X = x) = 1$$

Mean

$$E(X) = \sum xP(X = x)$$

Variance

$$V(X) = E(X^2) - [E(X)]^2$$

Standard deviation

$$\sigma(X) = \sqrt{V(X)}$$

Transformation

$$E(aX + b) = aE(X) + b \quad V(aX + b) = a^2V(X)$$

EGYPTIAN BACCALAUREATE • 2026

MATHEMATICS

CLASSROOM WORKING EDITION

Chapter 08

Statistical Inference

1 Concepts | 58 questions

B5 Landscape | one question per teaching page

Eng Eslam Ahmed

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01120009622

Chapter 08 | Statistical Inference

CLASSROOM ROUTE

Concept 2 | Expected Value and Variance of Multiple Variables

58 questions

Official lessons: 8-8, 8-9, 8-10, 8-11, 8-12

The Formula Bank changes at each Concept boundary.

Concept 2 | Expected Value and Variance of Multiple Variables — **OFFICIAL LESSONS 8-8, 8-9, 8-10, 8-11, 8-12**

Expected Value and Variance of Multiple Variables

Equation-first recall | use the same rail beside every question in this Concept

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$E(XYZ)=E(X)E(Y)E(Z)$ $\quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Start with the family cue, write the first move, then use the working grid.

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | V01

Q001 | WRITTEN

C08-K02-8-8-WU

Given the probability distributions of X and Y in the source tables, find E(X+Y).

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | V02

Q002 | WRITTEN

C08-K02-8-8-COIN

Three coins are tossed once; let X, Y and Z be the head indicators. Find E(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | V03

Q003 | WRITTEN

C08-K02-8-8-TRY

For the two source tables with X=1, 2, 3 and Y=4, 5, 6, find E(X+Y).

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | V04

Q004 | WRITTEN C08-K02-8-8-EX

For the two bags in the source exercise, find the expected total number of red balls drawn.

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | V05

Q005 | WRITTEN

C08-K02-8-8-CH

Three independent subsystems each receive two signals with success probability 0.5. Find E(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum

$E(X+Y)=E(X)+E(Y)$

Linear combination

$E(aX+bY)=aE(X)+bE(Y)$

Independent product

$E(XY)=E(X)E(Y)$

Variance of a sum

$V(X+Y)=V(X)+V(Y)$

Three variables

$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | V01

Q006 | WRITTEN

C08-K02-8-9-WU

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z.

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | V02

Q007 | WRITTEN

C08-K02-8-9-TRY

Two 500-point coins and two 100-point coins are tossed. Find EZ.

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | V03

Q008 | WRITTEN

C08-K02-8-9-DICE

A die is thrown twice and the score is Z=3X+5Y. Find EZ.

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | V04

Q009 | WRITTEN C08-K02-8-9-EX

For three 100-point coins and one 50-point coin, find the expected score from heads.

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | V05

Q010 | WRITTEN

C08-K02-8-9-CH

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | V01

Q011 | WRITTEN

C08-K02-8-10-WU

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find E(XY).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | V02

Q012 | WRITTEN C08-K02-8-10-TRY

For independent X with probabilities 2/5, 2/5, 1/5 at 1, 2, 3 and Y uniform on 2, 4, 6, find E(XY).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | V03

Q013 | WRITTEN

C08-K02-8-10-EX

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find E(XY).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | V04

Q014 | WRITTEN

C08-K02-8-10-CH

A card numbered 1 to 4 is replaced and a second card is drawn. Find E(XY).

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | V05

Q015 | WRITTEN

C08-K02-8-10-EX2

For independent X and Y with EX=1 and EY=2/7, find E(XY).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | V01

Q016 | WRITTEN C08-K02-8-11-WU

A card X is drawn from 1, 2, 3 and three-coin heads Y is independent. Find V(X+Y) and sigma(X+Y).

WORKING AREA

FORMULA BANK

Sum
$E(X+Y) = E(X) + E(Y)$

Linear combination
$E(aX+bY) = aE(X) + bE(Y)$

Independent product
$E(XY) = E(X)E(Y)$

Variance of a sum
$V(X+Y) = V(X) + V(Y)$

Three variables
$E(XYZ) = E(X)E(Y)E(Z)$ $V(X+Y+Z) = V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | V02

Q017 | WRITTEN

C08-K02-8-11-DICE

Two independent dice are thrown. Find the variance and standard deviation of their sum.

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | V03

Q018 | WRITTEN

C08-K02-8-11-EX1

An independent coin head count X and die score Y are combined. Find V(X+Y) and sigma(X+Y).

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | V04

Q019 | WRITTEN

C08-K02-8-11-EX2

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | V05

Q020 | WRITTEN

C08-K02-8-11-CH

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find V(X+Y) and sigma(X+Y).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | V01

Q021 | WRITTEN

C08-K02-8-12-WU

Three independent special dice take values 2, 4, 6 equally. Find E(XYZ) and V(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum

$$E(X+Y)=E(X)+E(Y)$$

Linear combination

$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product

$$E(XY)=E(X)E(Y)$$

Variance of a sum

$$V(X+Y)=V(X)+V(Y)$$

Three variables

$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | V02

Q022 | WRITTEN

C08-K02-8-12-TRY

Three independent dice take values 1, 3, 5 equally. Find E(XYZ) and V(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y) = E(X) + E(Y)$$

Linear combination
$$E(aX+bY) = aE(X) + bE(Y)$$

Independent product
$$E(XY) = E(X)E(Y)$$

Variance of a sum
$$V(X+Y) = V(X) + V(Y)$$

Three variables
$$E(XYZ) = E(X)E(Y)E(Z) \quad V(X+Y+Z) = V(X) + V(Y) + V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | V03

Q023 | WRITTEN

C08-K02-8-12-EX

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | V04

Q024 | WRITTEN

C08-K02-8-12-CH

Three friends have 5, 4 and 3 independent rolls to score a six. Find E(XYZ) and V(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | V05

Q025 | WRITTEN

C08-K02-8-12-EX2

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find E(XYZ).

WORKING AREA

FORMULA BANK

Sum

$E(X+Y)=E(X)+E(Y)$

Linear combination

$E(aX+bY)=aE(X)+bE(Y)$

Independent product

$E(XY)=E(X)E(Y)$

Variance of a sum

$V(X+Y)=V(X)+V(Y)$

Three variables

$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VMCQ

Q026 | MCQ

C08-K02-8-8-GM01

E(X+Y) equals Which of the following is correct?:

A EX-EY

B EX+EY

C EXEY

D EX/EY

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VMCQ

Q027 | MCQ

C08-K02-8-8-GM02

For three variables, E(X+Y+Z) equals Which of the following is correct?:

A EXEYEZ

B EX+EY+EZ

C EX-EY-EZ

D 0

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VMCQ

Q028 | MCQ

C08-K02-8-8-GM03

The expected heads in one fair coin toss is:

A 0

B 1/2

C 1

D 2

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VMCQ

Q029 | MCQ

C08-K02-8-8-GM04

For two signals with p=0.5, EX is:

A 0.5

B 1

C 2

D 4

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VMCQ

Q030 | MCQ

C08-K02-8-8-GM05

If EX=2 and EY=5, E(X+Y) is:

A 3

B 7

C 10

D 25

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VMCQ

Q031 | MCQ

C08-K02-8-9-GM06

E(aX+bY) equals Which of the following is correct?:

A aEX+bEY

B abE(XY)

C EX+EY

D $a+b$

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VMCQ

Q032 | MCQ

C08-K02-8-9-GM07

The expected score of one fair 100-point coin is:

A 25

B 50

C 100

D 200

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VMCQ

Q033 | MCQ

C08-K02-8-9-GM08

For a fair die, E(3X+5Y) with two rolls is:

A 14

B 21

C 28

D 35

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VMCQ

Q034 | MCQ   C08-K02-8-9-GM09

A fixed entry fee changes Which of the following is correct?:

A variance only

B the mean only

C neither mean nor variance

D standard deviation only

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VMCQ

Q035 | MCQ

C08-K02-8-9-GM10

The expected number of winners in 2 of 10 tickets with 2 winners is:

A 1/5

B 2/5

C 1/2

D 4/5

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VMCQ

Q036 | MCQ

C08-K02-8-10-GM11

E(XY)=EXEY requires Which of the following is correct?:

A $X = Y$

B independence

C zero means

D equal variances

WORKING AREA

FORMULA BANK

Sum
$E(X+Y) = E(X) + E(Y)$

Linear combination
$E(aX+bY) = aE(X) + bE(Y)$

Independent product
$E(XY) = E(X)E(Y)$

Variance of a sum
$V(X+Y) = V(X) + V(Y)$

Three variables
$E(XYZ) = E(X)E(Y)E(Z)$ $V(X+Y+Z) = V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VMCQ

Q037 | MCQ

C08-K02-8-10-GM12

For three fair coins, EX is:

A 1/2

B 1

C 3/2

D 3

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VMCQ

Q038 | MCQ

C08-K02-8-10-GM13

For a fair die, EY is:

A 2.5

B 3

C 3.5

D 6

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VMCQ

Q039 | MCQ

C08-K02-8-10-GM14

With replacement, two card draws are:

A dependent

B independent

C identical outcomes

D impossible

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VMCQ

Q040 | MCQ

C08-K02-8-10-GM15

If EX=2 and EY=3 for independent variables, E(XY) is:

A 5

B 6

C 1

D 9

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VMCQ

Q041 | MCQ — C08-K02-8-11-GM16

For independent X,Y, V(X+Y) equals Which of the following is correct?:

A VX-VY	**B** VX+VY
C VXVY	**D** 0

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VMCQ

Q042 | MCQ C08-K02-8-11-GM17

The variance of a fair die is:

A 7/2

B 35/12

C 35/6

D 1/4

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VMCQ

Q043 | MCQ

C08-K02-8-11-GM18

Sigma(X+Y) is:

A $V(X+Y)$

B sqrt$V(X+Y)$

C VX+VY

D $E(X+Y)$

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z)$ $V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VMCQ

Q044 | MCQ

C08-K02-8-11-GM19

The variance of the sum of two independent fair dice is:

A 35/12

B 35/6

C 70/6

D 49/4

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VMCQ

Q045 | MCQ

C08-K02-8-11-GM20

For independent sums, a constant shift changes Which of the following is correct?:

A variance

B standard deviation

C neither spread measure

D both spread measures

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VMCQ

Q046 | MCQ

C08-K02-8-12-GM21

For independent X,Y,Z, E(XYZ) equals Which of the following is correct?:

A EX+EY+EZ

B EXEYEZ

C 0

D E(XY)+EZ

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VMCQ

Q047 | MCQ C08-K02-8-12-GM22

For independent X,Y,Z, V(X+Y+Z) equals Which of the following is correct?:

A VXVYVZ

B VX+VY+VZ

C VX-VY-VZ

D 0

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VMCQ

Q048 | MCQ C08-K02-8-12-GM23

For a binomial count with n=5 and p=1/6, EX is:

A 1/6

B 5/6

C 5/36

D 6

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VMCQ

Q049 | MCQ C08-K02-8-12-GM24

For three values 2,4,6 equally likely, EX is:

A 2

B 3

C 4

D 6

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VMCQ

Q050 | MCQ

C08-K02-8-12-GM25

For three independent fair dice, V(sum) is:

A 35/12

B 35/6

C 35/4

D 343/8

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F01 | C08-K02-F01-VQUIZ

Quiz W1 | QUIZ WRITTEN

C08-K02-Q-W01

If EX=4 and EY=7, find E(3X-2Y).

WORKING AREA

FORMULA BANK

Sum
$E(X+Y)=E(X)+E(Y)$

Linear combination
$E(aX+bY)=aE(X)+bE(Y)$

Independent product
$E(XY)=E(X)E(Y)$

Variance of a sum
$V(X+Y)=V(X)+V(Y)$

Three variables
$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VQUIZ

Quiz W2 | QUIZ WRITTEN

C08-K02-Q-W02

For independent variables with EX=2 and EY=5, find E(XY).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VQUIZ

Quiz W3 | QUIZ WRITTEN

C08-K02-Q-W03

Two independent fair dice are thrown. Find V(X+Y) and sigma(X+Y).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VQUIZ

Quiz W4 | QUIZ WRITTEN

C08-K02-Q-W04

Three independent Binomial counts have (n, p)=(2, 1/2), (3, 1/2), (4, 1/2). Find V(X+Y+Z).

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F02 | C08-K02-F02-VQUIZ

Quiz M1 | QUIZ MCQ C08-K02-Q-M05

If $EX=3$ and $EY=4$, E(2X+Y) is:

A 7

B 10

C 14

D 24

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F03 | C08-K02-F03-VQUIZ

Quiz M2 | QUIZ MCQ

C08-K02-Q-M06

For independent variables EX=2, EY=5, E(XY) is:

A 7

B 10

C 25

D 1/2

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F04 | C08-K02-F04-VQUIZ

Quiz M3 | QUIZ MCQ C08-K02-Q-M07

For two independent die rolls, V(X+Y) is:

A 35/12

B 35/6

C 35/4

D 7/2

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$

Concept 2 | Expected Value and Variance of Multiple Variables

F05 | C08-K02-F05-VQUIZ

Quiz M4 | QUIZ MCQ | C08-K02-Q-M08

For three independent variables, V(X+Y+Z) is:

A the product of variances

B the sum of variances

C the sum of means

D always zero

WORKING AREA

FORMULA BANK

Sum
$$E(X+Y)=E(X)+E(Y)$$

Linear combination
$$E(aX+bY)=aE(X)+bE(Y)$$

Independent product
$$E(XY)=E(X)E(Y)$$

Variance of a sum
$$V(X+Y)=V(X)+V(Y)$$

Three variables
$$E(XYZ)=E(X)E(Y)E(Z) \quad V(X+Y+Z)=V(X)+V(Y)+V(Z)$$